

B.Tech. Degree I Semester Regular Examination November 2023

EE/EC/CS/IT 23-200-0102 B ENGINEERING PHYSICS
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Interpret modern devices and technologies based on lasers and optical fibres.
- CO2: Explain the basic principles of crystal physics.
- CO3: Summarise the characteristics and applications of superconducting materials, nano materials and smart materials.
- CO4: Illustrate the theory of semiconductors and magnetic materials.
- CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a)	Discuss the significance of Einsteins coefficients.	2	L2	1	1
(b)	Give two differences between characteristic and continuous X-rays.	2	L1	2	1
(c)	State any two applications of shape memory alloys.	2	L3	3	1
(d)	Define the magnetic susceptibility and describe how is it connected to the relative permeability.	2	L2	4	1
(e)	How do you relate the fermi level to the conductivity of semiconductors?	2	L1	4	2

PART B

(4 × 10 = 40)

II.	(a)	Describe the apparatus, principle and working of a four level laser with suitable diagrams.	7	L2	1	1
	(b)	Spontaneous emission and stimulated emission should be equal for a laser output. Is this statement true or false? Justify your answer.	3	L4	1	2
OR						
III.	(a)	Describe the propagation technique in step index and graded index optical fibres with diagrams.	7	L2	1	1
	(b)	Find the angle of refraction at the medium core interface and the critical angle of an optical fibre of core refractive index 1.54 and cladding refractive index 1.45.	3	L3	1	2

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) What are Miller indices? Determine the Miller indices of a plane parallel to the z axis and having intercepts of 2 and 2/3 along x and y axes of a cubic crystal system. Give any two of its characteristics.	7	L2	2	1
	(b) Draw the (100), (211) and (040) planes for a unit cell of the simple cubic structure.	3	L2	2	2
OR					
V.	(a) Explain the principle and working of the x-ray diffractometer.	7	L2	2	1
	(b) Calculate the lattice constant of a substance with fcc structure, molecular weight of 60.2 and density of 6250 kg/m ³ .	3	L3	2	2
OR					
VI.	(a) Explain the basis of classification in superconductors and the differences with relevant diagrams.	7	L1	3	1
	(b) Prove that a material in the superconducting state is a perfect diamagnet.	3	L3	3	1
OR					
VII.	(a) Describe in detail the working of a twisted nematic liquid crystal display system.	7	L2	3	1
	(b) The critical temperature of a superconducting sample with atomic mass of 26.91 is 1.2 K. Estimate the critical temperature when the isotopic mass changes to 32.12.	3	L3	3	2
OR					
VIII.	(a) Provide a comparative study between paramagnetic, diamagnetic and ferromagnetic materials.	7	L1	4	1
	(b) Find the percentage increase in magnetic induction when the space within a current carrying toroid is filled with magnesium with susceptibility of 1.2×10^{-3} .	3	L3	4	2
OR					
IX.	(a) On the basis of band gap theory, describe semiconductors and their properties.	7	L1	4	1
	(b) Give any three specific applications of semiconductors.	3	L3	4	2

Blooms's Taxonomy Levels

L1 - 24%, L2 - 38%, L3 - 33%, L4 - 5%.
